

Imports

```
In [197...import pandas as pd # for working with data
import numpy as np # for working with data
import seaborn as sns # for making visualizations
from matplotlib import pyplot as plt # for making visualizations
from sklearn.model_selection import cross_validate
from sklearn import metrics
import matplotlib.pyplot as plt
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
digits = load_digits()
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test=train_test_split(X,y,test_size=0.2)
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB, GaussianNB
from sklearn.feature_extraction.text import CountVectorizer
```

```
In [198...import chardet
with open('spam.csv', 'rb') as rawdata:
    result = chardet.detect(rawdata.read(100000))
result
df = pd.read_csv('spam.csv',encoding='Windows-1252')
df.head()
```

Out[198]:

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

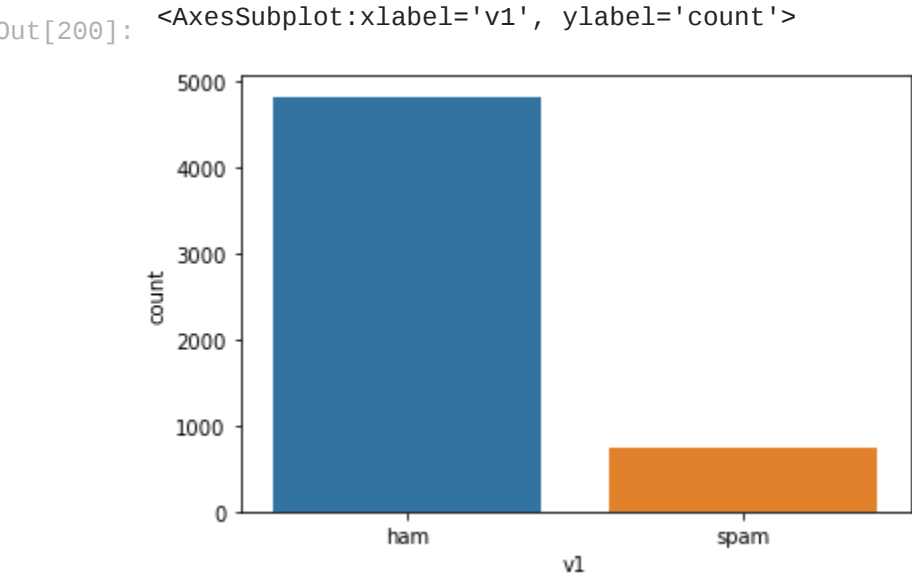
Pre-processing

```
In [199...df.isna().sum()
```

Out[199]:

```
v1      0
v2      0
Unnamed: 2    5522
Unnamed: 3    5560
Unnamed: 4    5566
dtype: int64
```

```
In [200...sns.countplot(x=df["v1"])
```



```
In [201...df['v1'].value_counts()
```

Out[201]:

```
ham      4825
spam      747
Name: v1, dtype: int64
```

```
In [202...df['v2'].value_counts()
```

Out[202]:

```
Sorry, I'll call later
30
I cant pick the phone right now. Pls send a message
12
Ok...
10
7 wonders in My WORLD 7th You 6th Ur style 5th Ur smile 4th Ur Personality 3rd Ur Nature 2nd Ur SMS and 1st \Ur Lovely Friendship\"... good morning dear"
4
Say this slowly.? GOD,I LOVE YOU & I NEED YOU,CLEAN MY HEART WITH YOUR BLOOD.Send this to Ten special people & u c miracle tomorrow, do it,pls,pls do it...
4

..
I gotta collect da car at 6 lei.
1
No. On the way home. So if not for the long dry spell the season would have been over
1
Urgent! Please call 09061743811 from landline. Your ABTA complimentary 4* Tenerife Holiday or â£5000 cash await collection SAE T&Cs Box 326 CW25WX 150ppm
1
Dear 0776xxxxxxx U've been invited to XCHAT. This is our final attempt to contact u! Txt CHAT to 86688 150p/MsgrcvdHG/Suite342/2Lands/Row/W1J6HL LDN 18yrs
1
Rofl. Its true to its name
1
Name: v2, Length: 5169, dtype: int64
```

```
In [203...# df = pd.get_dummies(df, columns = ['v1', 'v2'])
df['v1'] = df["v1"].map({'spam':1,'ham':0})
```

```
In [204...df = df.drop(columns = ['Unnamed: 2', 'Unnamed: 3', 'Unnamed: 4'])
df
```

Out[204]:

	v1	v2
0	0	Go until jurong point, crazy.. Available only ...
1	0	Ok lar... Joking wif u oni...
2	1	Free entry in 2 a wkly comp to win FA Cup fina...
3	0	U dun say so early hor... U c already then say...
4	0	Nah I don't think he goes to usf, he lives aro...
...	...	...
5567	1	This is the 2nd time we have tried 2 contact u...
5568	0	Will i_b going to esplanade fr home?
5569	0	Pity, * was in mood for that. So...any other s...
5570	0	The guy did some bitching but I acted like i'd...
5571	0	Rofl. Its true to its name

5572 rows × 2 columns

```
In [205...df.isnull().sum()
```

Out[205]:

```
v1      0
v2      0
dtype: int64
```

```
In [211...words = ['free', 'money', 'win money', 'win', 'you won', 'congratulations you', 'call now', 'cash', 'contact now']
vectorizer = CountVectorizer(stop_words='english')
count_matrix = vectorizer.fit_transform(words)
count_array = count_matrix.toarray()
df1 = pd.DataFrame(data = count_array,columns = vectorizer.get_feature_names_out())

df1
```

Out[211]:

	cash	congratulations	contact	free	money	win	won
0	0	0	0	1	0	0	0
1	0	0	0	0	1	0	0
2	0	0	0	0	1	1	0
3	0	0	0	0	0	1	0
4	0	0	0	0	0	0	1
5	0	1	0	0	0	0	0
6	0	0	0	0	0	0	0
7	1	0	0	0	0	0	0
8	0	0	1	0	0	0	0

```
In [212...X = vectorizer.fit_transform(df["v2"])

y = df["v1"]
X_train, X_test, y_train, y_test = train_test_split(X.toarray(), y, test_size=0.2, random_state=1)
```

```
In [214...clf = GaussianNB()

clf.fit(X_train, y_train)

clf.score(X_test, y_test)
```

Out[214]:

```
0.8977578475336323
```

In []:

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In []:

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In []: